4.1.1.3 Specific heat capacity

In this practical you will measure the temperature changes of different materials when they are heated so you can calculate their specific heat capacity. This investigation involves linking the decrease of one energy store (or work done) to the increase in thermal energy stored. The energy transferred (work done) will cause a temperature rise.

In this investigation, the temperature rise of a material depends on its **specific heat capacity**. Materials with a low specific heat capacity (a low capacity to store thermal energy) will have a greater temperature increase than those with a high specific heat capacity.

Learning outcomes	Maths skills required	Formulae
• Safely collect data to calculate the specific heat capacity of a metal block, or of water. • Understand how to identify any anomalous results.	• Substitute numerical values into algebraic equations using appropriate units for physical quantities. • Plot two variables from experimental data. • Determine the slope of a linear graph.	• change in thermal energy (J) = mass (kg) × specific heat capacity (J/kg °C) × change in temperature (°C) $\Delta E = mc\Delta\theta$ • energy transferred (J) = power (W) × time (s) power (W) = • potential difference (V) × current (A)

Apparatus list

- copper block with two holes, for a thermometer and a heater
- aluminium block with two holes, for a thermometer and a heater
- 250 cm^3 beaker
- water
- thermometer
- petroleum jelly
- 50 W, 12 V heater
- 12 V power supply
- insulation to wrap around the block or beaker
- ammeter
- voltmeter
- five leads
- stopwatch
- balance (mass)
- eye protection

Safety notes

- Wear eye protection at all times.
- Be careful with water around electricity.
- The heating element will get very hot, especially if it is not inside a metal block. Take care not to burn yourself.
- If any of the equipment is damaged do not use it.
- If you scald yourself with hot water, cool the burn under cold running water immediately and ask your teacher for assistance.

Common mistakes

- The heating element should fit very snugly into the metal block, but there may be a small layer of air between the heating element and the metal block. Add a drop of water before you put the heating element in to improve transfer of energy between the heating element and the metal block.

- Remember to measure the mass of the metal block. These blocks are usually 1 kg, but to make sure your calculations are accurate, you should take an accurate mass measurement.
- Your teacher might tell you the power of your heater. You can trust your teacher, or you can attach an ammeter and voltmeter and calculate the power using:

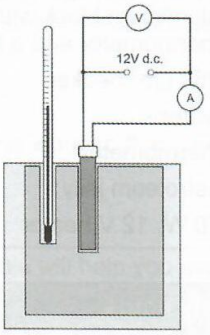
  power (W) = potential difference (V) × current (A)
  $P = VI$
- Make sure you heat the metal block for at least 10 minutes; otherwise you will not be able to draw a graph with a good range of results.
- Don't forget to use your graph to find the gradient of the line. You will need this and the mass of the block to work out the specific heat capacity of your metals.

Method

Read these instructions carefully before you start work.

1. Choose your material, for example, a copper or aluminium block, and measure its mass, in kilograms. (Note: if choosing water, first find the mass of the beaker, then the mass of the beaker and the water, then subtract the mass of the beaker to determine the mass of the water.)
2. Wrap insulation around your block or beaker.
3. Smear petroleum jelly around the bulb end of the thermometer (not necessary if measuring temperature of water rather than of metal block) then put the thermometer in the small hole in the block (or into the water).
4. Measure the starting temperature of the block (or water).
5. Put a heater in the larger hole in the block. Connect the ammeter, power pack and heater in series, as shown in **Figure 1**.
6. Connect the voltmeter across the power pack, as shown in **Figure 1**.
7. Turn the power pack to 12 V and switch it on. Start the stopwatch as you turn on the power pack.
8. Record the ammeter and voltmeter readings every 60 seconds in **Table 1**. These may vary slightly during the experiment, but not significantly. Record the temperature of the metal block (or water) every 60 seconds in **Table 2**. If you are using water, stir it at regular intervals.

Figure 1



9. After 10 minutes, turn off the power pack.
10. Keep the thermometer in the metal block (or water) for a while longer. Record the **maximum** temperature of the metal block (or water) in **Table 2** – this may be a little while after you have turned off the power pack.
11. Calculate the power of the heater and record the values in **Table 1**. Then calculate the energy transferred (work done) up to each time point. Use the following equations:
    power (W) = potential difference (V) × current (A)
    energy transferred (J) = power (W) × time (s)
12. Plot your results on **Graph 1**, with temperature change (°C) on the *y*-axis and energy transferred (J) on the *x*-axis.

Record your results

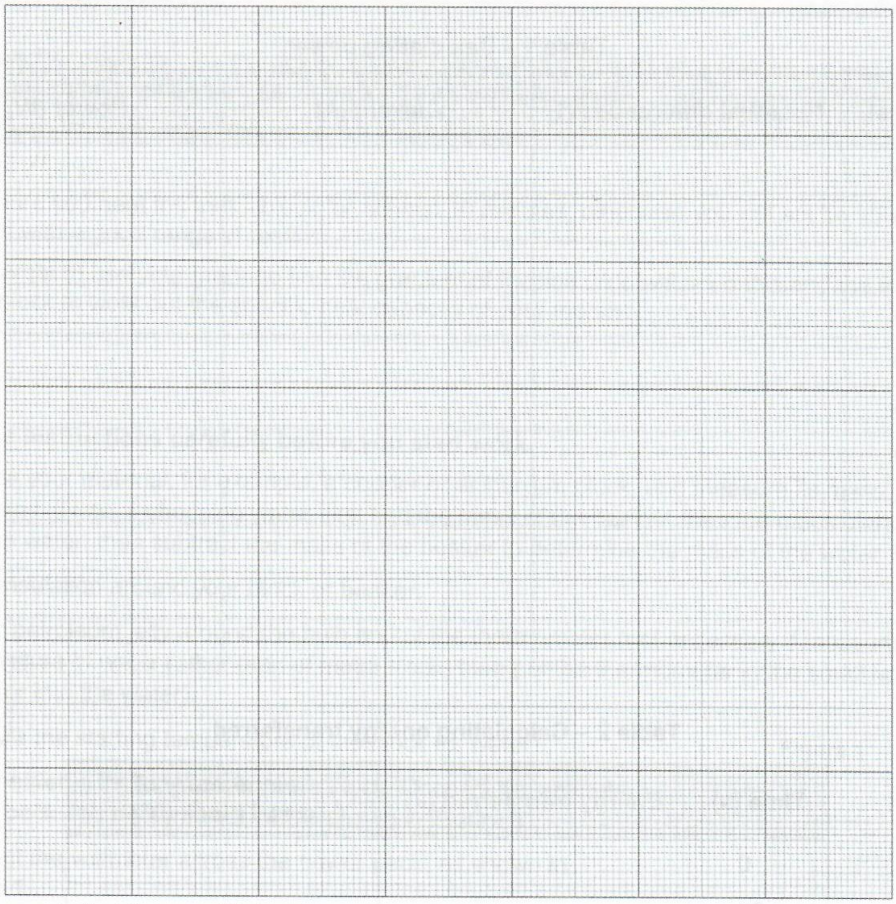
Table 1 – Calculating power

Time (s)	Potential difference (V)	Current (A)	Power (W)
0			
60			
120			
180			
240			
300			
360			
420			
480			
540			
600			

Table 2 – Calculating energy transferred

Time (s)	Energy transferred (J)	Temperature of metal block (°C)
0		
60		
120		
180		
240		
300		
360		
420		
480		
540		
600		
Maxiumum temperature of block		

Graph 1



Theory suggests that the specific heat capacity can be found from:

$$\text{specific heat capacity} = \frac{1}{\text{mass} \times \text{gradient}}$$

Use your graph and this equation to determine the specific heat capacity of your metal.

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